

Nahamcon 2020 Crypto - December

deuterium — 2020-06-14

Canonical HTML: <https://deut-erium.github.io/WriteUps/2020/nahamconCTF/crypto/December/2020-06-14-Nahamcon-2020-December.html>

Abstract

CTF writeup: Four weak DES keys audition for NahamCon 2020's December. OFB decryption gives one the part.

December

Challenge

105 Solves

×

December
125

This is my December...

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📄 source.py

📄 ciphertext

Flag

Submit

We are provided with source.py, which reads

```
#!/usr/bin/env python

from Crypto.Cipher import DES

with open('flag.txt', 'rb') as handle:
    flag = handle.read()
```

```
padding_size = len(flag) + (8 - ( len(flag) % 8 ))
flag = flag.ljust(padding_size, b'\x00')

with open('key', 'rb') as handle:
    key = handle.read().strip()

iv = "13371337"
des = DES.new(key, DES.MODE_OFB, iv)
ct = des.encrypt(flag)

with open('ciphertext', 'wb') as handle:
    handle.write(ct)
```

And a binary ciphertext

```
00000000: d6a2 6fe5 c75c 22e0 5413 5e4e 1140 5d58  ..o..\".T.^N.@]X
00000010: f5ea 69f9 d419 31f7 5513 5745 5452 5e44  ..i...1.U.WETR^D
00000020: 889e 62ff d15c 2ae1 115e 5617 4141 5643  ..b..\*...^V.AAVC
00000030: e7a4 6eff cc1b 49f4 5d52 544c 455b 5a44  ..n...I.]RTLE[ZD
00000040: dda3 79c9 c310 2fcd 586c 5d52 5457 4e37  ..y.../.Xl]RTWN7
```

Which is quite unusual in the sense that most of it is readable in unusual sizes of 8.

This could be another instance of Weak keys

Lets quickly run through the 4 weak keys

```
m Crypto.Cipher import DES
with open('ciphertext', 'rb') as ct_file:
    ct = ct_file.read()

weak_keys = [
    b'\x01\x01\x01\x01\x01\x01\x01\x01',
    b'\xfe\xfe\xfe\xfe\xfe\xfe\xfe\xfe',
    b'\xe0\xe0\xe0\xe0\xf1\xf1\xf1\xf1',
    b'\x1f\x1f\x1f\x1f\x0e\x0e\x0e\x0e'
]

for key in weak_keys:
    iv = b"13371337"
    des = DES.new(key, DES.MODE_OFB, iv)
    pt = des.decrypt(ct)
    print(pt)
```

Which produces the plaintexts

```
b'\xaf\xe0\xda\x8c=\xedG\xe4e my sno\x8c\xa8\xdc\x90.\xa8T\xf3d
dreams\xf1\xdc\xd7\x96+\xed0\xe5 me
pret\x9e\xe6\xdb\x966\xaa,\xf0lag{this\xa4\xel\xcc\xa09\xa1J\xc9i_need}\x00'
b'\xb8(\xff\x82\xb0)\x06$e my sno\x9b\xf9\x9e\xa3l\x153d
dreams\xe6\x14\xf2\x98\xa6)\x0e% me
pret\x89.\xfe\x98\xbbnm0lag{this\xb3)\xe9\xae\xb4e\x0b\ti_need}\x00'
b'c\xad\xc7\x9dVR$\xb2e my sno@\xe5\xc1\x81E\x177\xa5d dreams=\x91\xca\x87@R,\xb3 me
pretR\xab\xc6\x87]\x150\xa6lag{thish\xac\xd1\xb1R\x1e)\x9fi_need}\x00'
```

```
b'These are my snow covered dreams\nThis is me\npretending\nflag{this_is_all_i_need}\x00'
```

flag—this is all i need”

How to solve the challenge if not aware of weak keys?

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des otb

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Decrypt the message, find the flag, and then marvel at how broken everything is.

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